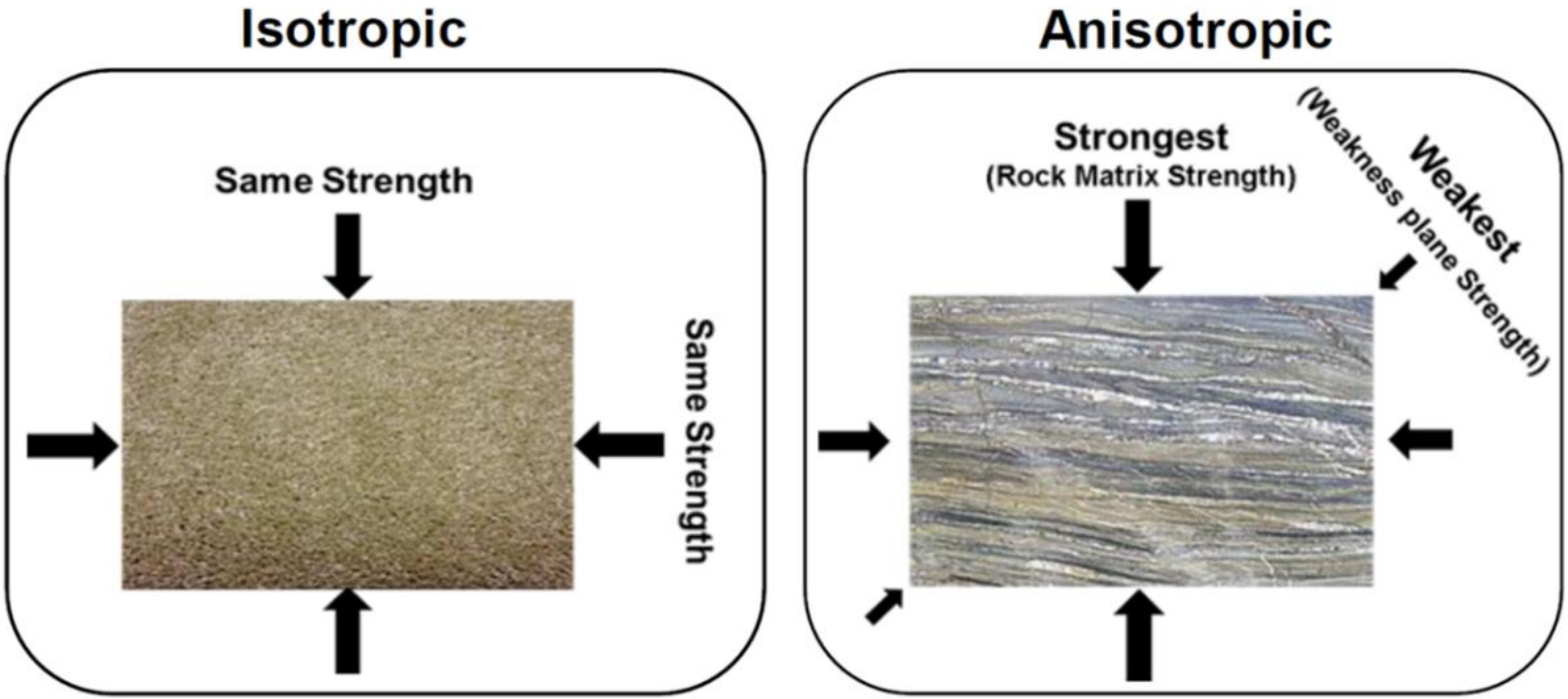
MATERIALS - AM

Anisotropic and Isotropic Properties

- Isotropy, which is defined as uniformity in all orientations. Thus, if we apply it to additive manufacturing, it refers specifically to uniformity in 3D printed parts. We can interpret this term from many different angles, whether it is the color, surface finish, shape of the model, etc.
- However, the property that most often interests experts, and to which the concept of isotropy is generally associated, is strength. In other words, an isotropic part is one that has the same strength in all the physical parts that make it up.
- In general, most thermoplastic polymers are considered isotropic. This is mainly due to their nature and the fact that the cohesion between their component polymer chains is uniform in all directions. However, this does not mean that parts made from these materials maintain this quality level.


Isotropic
Same Strength
Same Strength
Anisotropic
Strongest
(Rock Matrix Strength)
Weakest
(Weakness plane Strength)

Isotropy	Anisotropy
The properties of a material are identical with different directions i.e. they are independent of direction. The structure of the minerals is uniform (amorphous). This type of behaviour is called isotropy.	The properties of a material vary with different directions i.e. they are dependent on direction. The structure of the minerals is not uniform (crystalline). This type of behaviour is called anisotropy.
Isotropic minerals do not affect the polarization direction of the light. The velocity of light is same in all directions and do not have double refraction.	Anisotropic minerals affect the polarization direction of the light. With different velocity of light and have double refraction.
Isotropic mineral can appear dark or stay dark when light passes through it. The uniform structure of the mineral blocks the light in all directions.	Anisotropic minerals will allow some light to pass, and thus will be generally light, unless in specific orientations (as their structure is not uniform).
Isotropic minerals have same and consistent chemical bonding.	Anisotropic minerals have different and inconsistent chemical bonding.

- In fact, sometimes the manufacturing method can cause anisotropy in the parts even though their raw material has isotropy.
- When this happens and anisotropic parts are obtained, it means that they have physical properties that are different depending on the direction in which they are measured.

How To Achieve Isotropy in 3D Printing

- The importance of design and slicing
- Choosing a manufacturing process
- Post processing techniques

The importance of design and slicing

- The critical design and modeling stage is done at the beginning of any additive manufacturing process. Therefore, it is also an important step to take into account when it comes to achieving isotropic final parts.
- The simpler and more symmetrical the parts are, the easier it will be for their strength to be the same everywhere. And we are not only talking about the general shape but also about the inside of the parts. The design of the geometry itself can also have an influence since lattice-type structures defined according to requirements can replace solid volumes
- Finally, it is not only the geometrical conditions, the technology and the printing material that must be taken into account in this initial prefabrication phase. It is also important to understand the slicing process carried out on the slicer. These parameters can be adjusted in the slicer program and can guarantee a correct manufacturing process for the final parts.

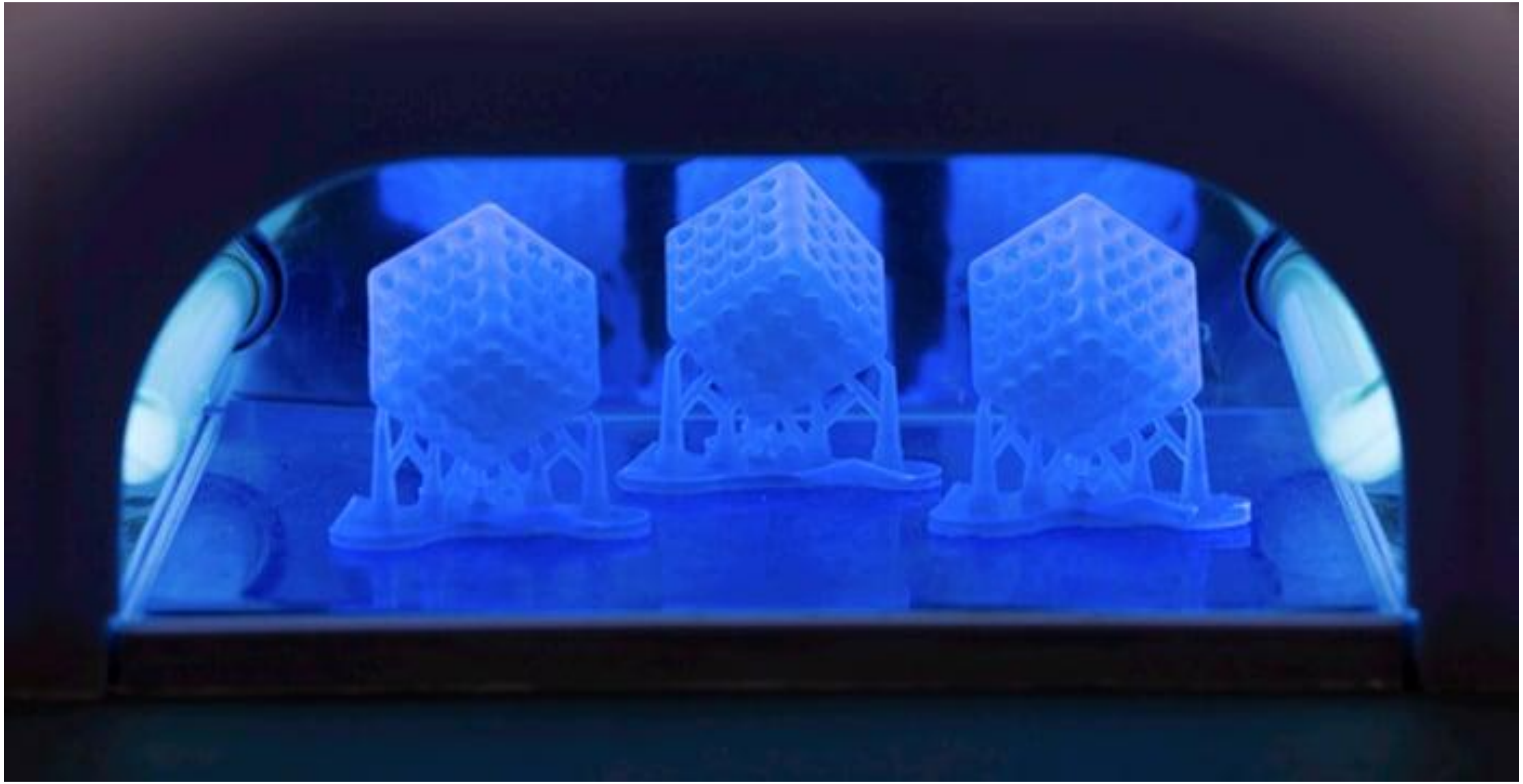
Choosing a manufacturing process

- A key point to take into account if we want to obtain isotropic parts is to choose the right additive manufacturing technology.
- If we look at FDM 3D printing, one of the most widely used techniques today, we find a multitude of problems related to isotropy.
- During the thermoplastic extrusion process, an intertwining of the molten polymer chains is generated between each of the adjacent layers in order to hold the part together.
- As a result, microporous structures may be generated, where the real cross-section of the part varies in each direction with respect to the apparent cross-section. Since these polymer chain bonds are not very strong, the models obtained will be weaker in relation to the lines of the layers.

- In resin 3D printing , individual monomers are covalently bonded by the application of a light source, resulting in the formation of a solid cured polymer layer. Typically, the exposed layer is not completely cured and still contains unbound monomer groups. When the next layer is exposed, it is possible to establish additional covalent bonds between the current layer and the previous one. These bonds contribute to the formation of a much stronger network structure and a closer approximation of isotropy compared to FDM technology.
- With SLS 3D printing, parts with ideal characteristics for engineering applications can be more easily obtained, since they have high isotropy, high dimensional accuracy and allow the creation of models with complex geometries. It should be noted that the models manufactured via SLS 3D printing may still have porosity in their structure. However, unlike those created using FDM, this porosity is uniform and is not influenced by the orientation of the part during the printing process.

Post processing techniques

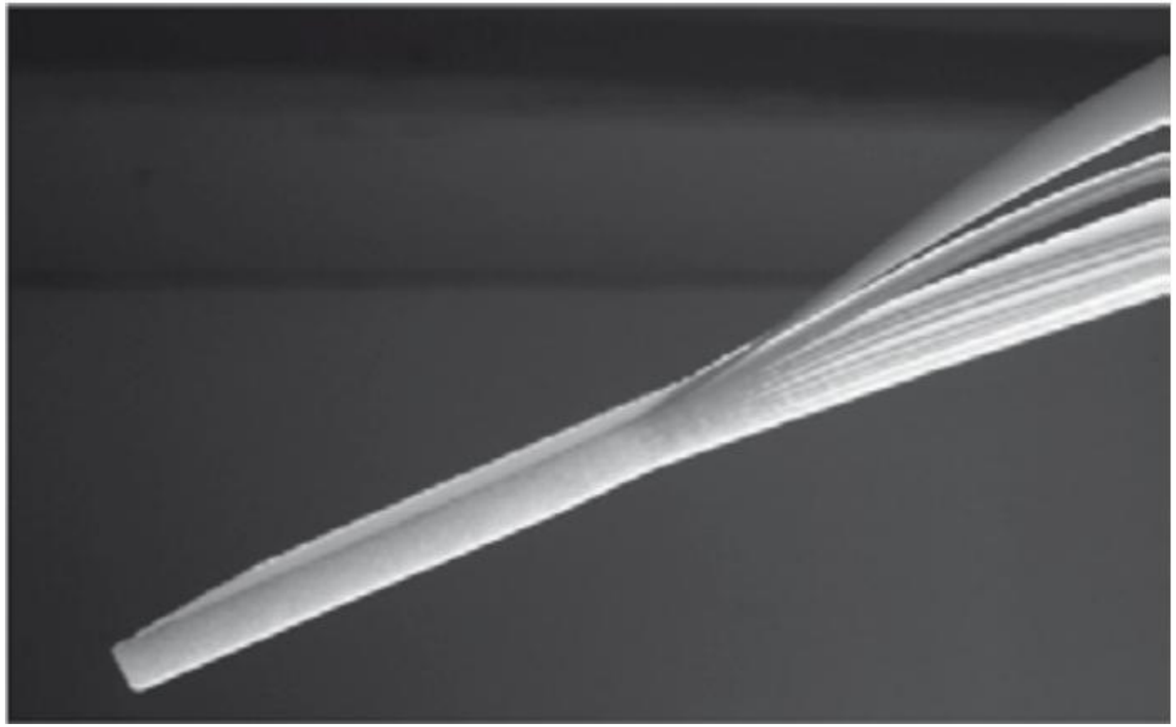
- Post-processing is the final necessary step in any additive manufacturing process. In this context, it can play a key role in increasing the bond between layers and reinforcing the isotropy of the parts.
- In the case of resin 3D printing, post-treatment techniques where heat is applied, such as thermal post-curing, promote further curing of the layers throughout the part.
- In addition, in cases where print support structures have been used, the removal of these may leave some defects on the surface of the parts. That is why some post-processing methods, such as polishing and sanding, can help to smooth these imperfections and avoid deformation of the parts for better isotropy.

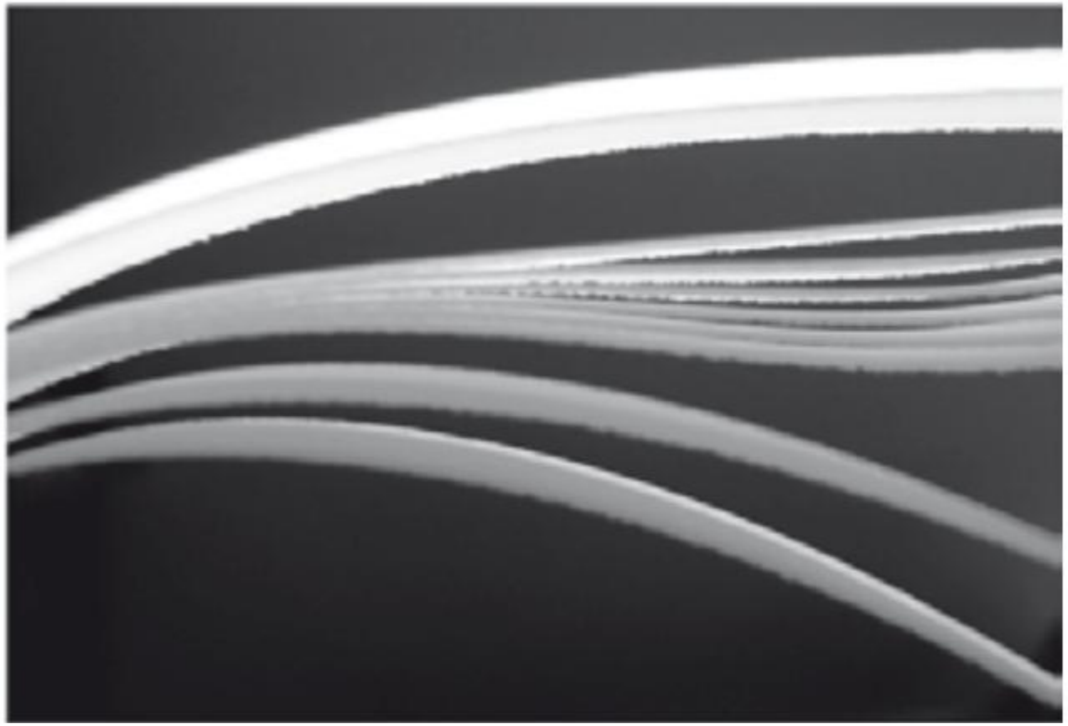


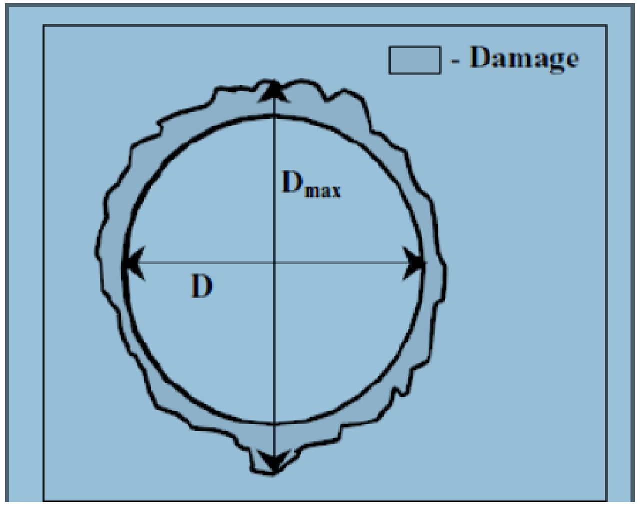
Problems with Anisotropic effect

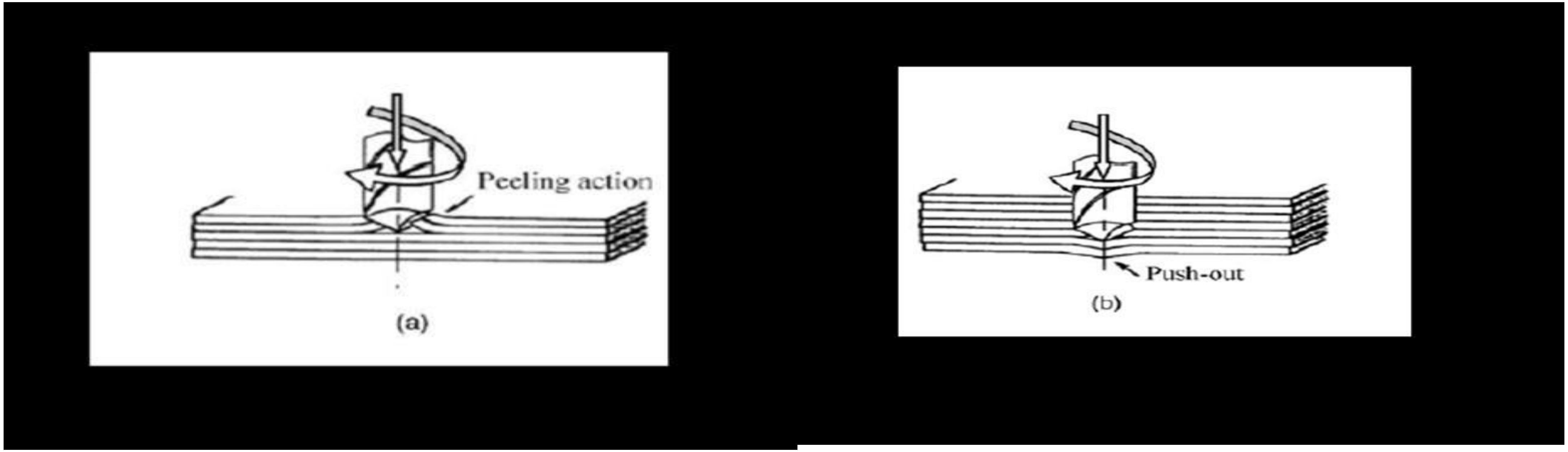
- The anisotropic effect is also closely linked to the way adjacent layers are bonded. The worst imaginable case is a delamination of layers. It is obvious that this might happen with FLM processes but delamination may occur with all AM processes.
- Similar, but again more pronounced, anisotropic effects result from 3D-printing processes (powder-binder-processes). However, here the porosity of the part has a bigger effect on the part’s properties than the layer-induced anisotropy. Infiltration reduces this effect, but does not eliminate it.

Delaminating of layers due to inaccurate build parameters set on purpose; laser sintering, polyamide






- Damage
D_{max}
D


Peeling action
(a)
Push-out
(b)

DRILLING INDUCED DELAMINATION

- Drilling is difficult due to their anisotropic, high strength, hard reinforced, high abrasive nature.

- It is an inter-ply failure phenomenon induced by drilling.

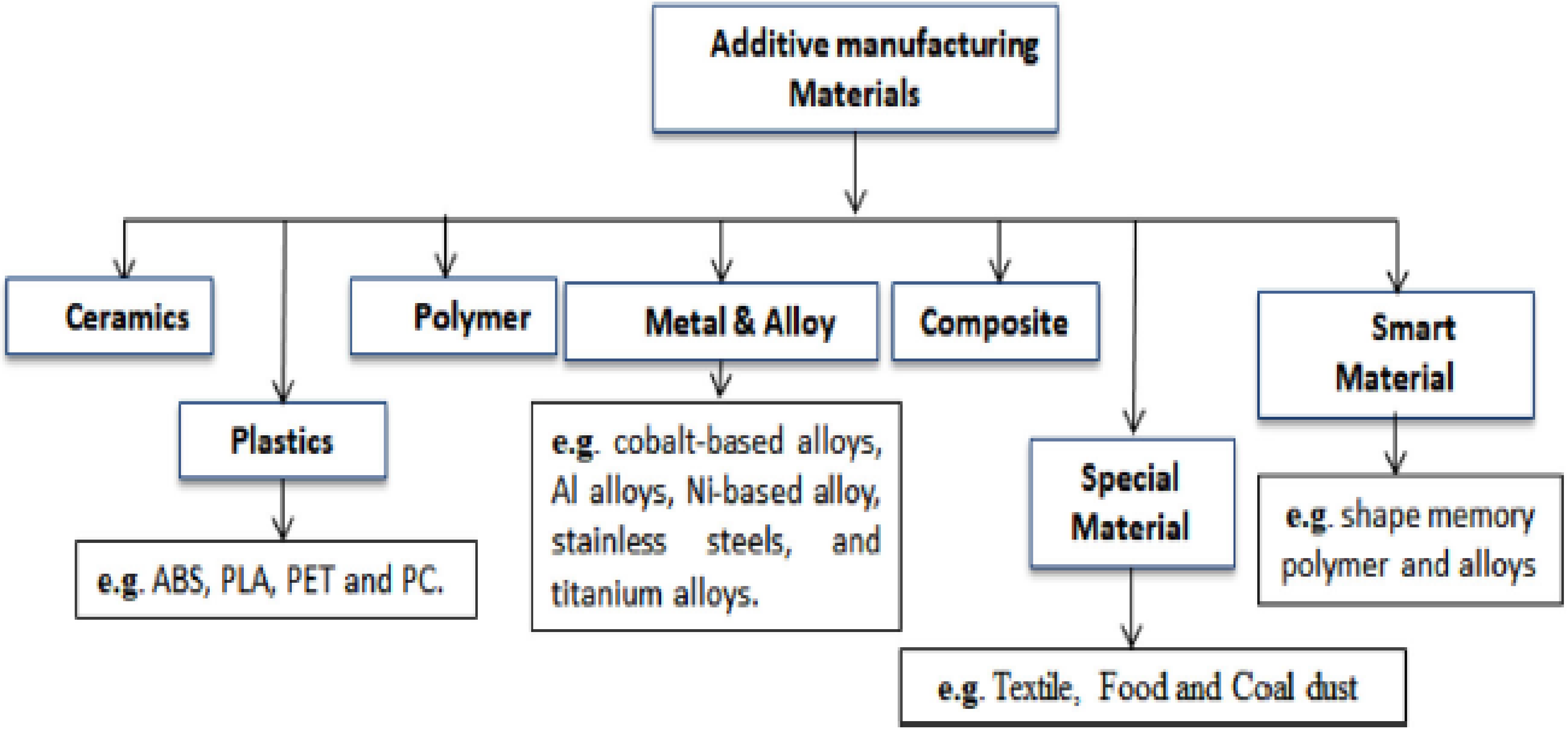
- It reduces strength and quality of material

- Peel-up and push-out delamination

- Compared to stereolithography, the extrusion processes (fused deposition modeling) show higher anisotropic effects. The process requires a pasty state while extruding the material, but does not allow the total melting in order to guarantee the geometrical stability of the partially finished build.
- This effect can be reduced by improved machine technology, especially by proper heat treatment and by using thin layers.
- Even the layer laminate manufacturing process, that basically shows the most elaborated anisotropic behavior, performs very heterogeneously. If paper is used and bonded by some kind of glue, the material is anisotropic.
- The anisotropic effect decreases, if plastic foils are used and are bonded by solvents.

- With direct manufacturing processes, anisotropic part behavior must be compensated in the part design phase and also when the build parameters for AM are set.
- This requires information about the directional differences of the material properties. In many cases, this information is not readily available and the proper calculation of the part is subject to experience.

- As a consequence, AM materials are usually developed by or under the responsibility of the machine manufacturer who treats the material as a proprietary product and exclusively sells it to his machine customers. Some users are skeptical, mainly because of economical reasons; but on the other hand, this approach guarantees a proper build.
- The continuous increase of overall material consumption forces the activities of so-called third-party suppliers that already entered the AM business. Mainly for materials for plastic laser sintering and stereolithography independent markets are developing.


Additive manufacturing Materials
Ceramics
Plastics
Polymer
Metal & Alloy
Composite
Special Material
Smart Material
e.g. ABS, PLA, PET and PC.
e.g. cobalt-based alloys, Al alloys, Ni-based alloy, stainless steels, and titanium alloys.
e.g. Textile, Food and Coal dust
e.g. shape memory polymer and alloys

Plastics

- Plastics were the first group of materials to be processed by AM and they still provide the biggest part of materials. Materials for stereolithography are acrylic or epoxy resins that must support photo polymerization.
- Today, the sticky and brittle materials of the early 1990s are replaced by materials that mimic materials for plastic injection molding. This was achieved by filling the resin with nano-particles to increase heat deflection temperature and mechanical stability.
- In addition, the variety of materials was increased and now includes transparent and non-transparent, elastic, stiff, and many more different materials.

ABS [Acryonitrile Butadiene Styrene]

- Acrylonitrile Butadiene Styrene (ABS) is a thermoplastic polymer widely used in the industry as it is recognized for its low-temperature impact resistance and ability to create lightweight parts.
- ABS filament is one of the most popular 3D printer materials. Elastomers are used to make the material more flexible and shock-resistant.
- ABS is a rigid material that can endure temperatures ranging from − 4°F to 176°F.
- It is a recyclable material that may be employed in chemical processes and has great strength. On the other hand, ABS is not a biodegradable substance and can shrink when exposed to moist air. It cannot be utilised in open-platform 3D printers; the printing platform must be closed and heated to avoid warping and shrinking.

ABS [Acryonitrile Butadiene Styrene]-Applications

- Kitchen appliance housings
- Automobile dashboard and bumper
- Protective equipment like headgear, electrical casings and covers
- Lego bricks and other toys
- Musical instruments

Polyamide (Nylon)

- Nylon, or Polyamide, is a thermoplastic synthetic substance frequently called plastic.
- Nylon is a low-cost thermoplastic material and one of the world’s strongest. It is used to make highly intricate geometrical structures.
- Nylon demands a high temperature to print due to its great flexibility, durability, low friction, and corrosion resistance properties, and it is more challenging to get Nylon to attach to the print bed than other materials.
- Some FDM 3D printers may be unsuitable for such high temperatures, as the average printing temperature for nylon filaments is 240° and above. Before utilizing nylon filaments, it is a good idea to double-check the extruder’s maximum temperature on the FDM printer.
- Selective Laser Sintering (SLS) or Fused Deposition Modelling (FDM) and Fused Filament Fabrication (FFF) printers primarily employ nylon.

Polyamide (Nylon)- Applications

- Ergonomic Tools
- Assembly Trays
- Cosmetic Parts

PLA [Polylactic Acid]

- PLA is a biodegradable substance that is also known as polylactic acid.
- It is made using organic ingredients like cornstarch.
- This substance is classified as a pseudoplastic non-Newtonian fluid. This means that its viscosity (flow resistance) will alter due to its exposure to stress.
- PLA is one of the most accessible 3D printing materials because it does not require a heated printing platform and prints at a lower temperature than required for printing ABS parts. Changing the design with such a fast cooling and solidification speed is tough.

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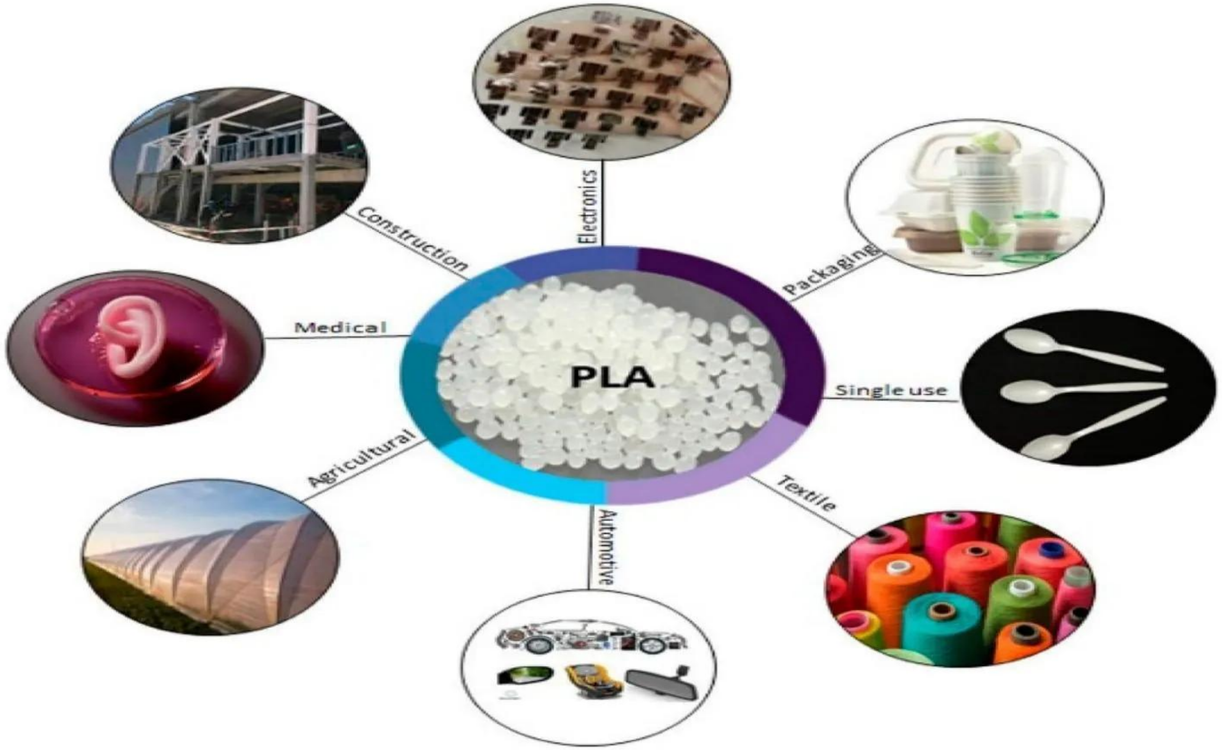
PLA [Polylactic Acid]

- Also, they can deteriorate if the PLA models come into touch with water.
- PLA is the most used material in FLA 3D printing because of its simplicity, colour variety, and characteristics. In addition, PLA offers a variety of alternatives due to its biodegradability, including mechanical recycling, chemical recycling, landfilling, and industrial composting

Applications

- Production of tubes
- Food packaging
- Cutlery
- Surgical threads



Photopolymer Resin

- One of the most popular 3D printing materials is resin. Resins are UV light-sensitive, and they solidify with the help of a light source or laser. Photopolymer resins are utilised in SLA, DLP multijet technologies. Intricate features with smoother surfaces are created with resins. The difference between FDM filament and resins is that one must work on mixing and matching resins to achieve different effects. Resins come in a variety of varieties to meet a variety of requirements.

Applications

- Jewellery making
- Dentistry
- Animation and props
- Miniatures
- Prototyping and product development

Polyether-ether-ketone (PEEK)

- PEEK is a high-quality organic thermoplastic polymer that is colourless and can model accurate, high-quality items. Semi-crystalline thermoplastics with outstanding mechanical qualities make up these materials. It has outstanding strength, dimensional stability, and stiffness and can withstand high temperatures, making it a viable alternative to metal. It has high wear resistance and a low coefficient of friction since it does not require lubrication. This material can be used where a constant high temperature is required, with a melting point of 343 °C. This material is suited for low-volume production and speciality designs where prototyping with metal and traditional processes is challenging.

Applications

- Medical implants, Piston parts, Bearing, Pumps, Electrical cable insulation, Compressor plate valves

Polyetheretherketone (PEEK)

Alumide

- Alumide is a blend of Polyamide (SLS) and Aluminium powder used in 3D printing.
- While Polyamide has the attributes of stiff plastic, Aluminium has a matte and metallic appearance, is slightly porous, and can withstand high temperatures. Because support structures are rarely required, and users can produce complex designs. It is mechanically strong and thermal resistant (up to 170 °C).
- Due to its durability and affordability, it provides excellent versatility for 3D printing, especially when it comes to complex models. It can therefore be used for mechanical parts with mild stresses or aesthetic pieces that need to look metallic.
- • Jewellery • Automotive parts • Decorative items • Cases casings to gadgets


Alumide

Stainless steel

- Stainless steel is well-known for its incredible strength and resistance to corrosion. Because of its remarkable physical strength, stainless steel offers a low-cost form of metal printing ideal for large products.
- It can also be utilised as a metal 3D printing material for complicated designs that are now impossible to produce using traditional manufacturing processes. Stainless steel is employed in various industries, including manufacturing and assistive technology. Fusion and laser sintering are the 3D printing processes utilised to create stainless steel. In the same way, as gold and silver are printed, stainless steel is printed using DMLS or Direct Metal Laser Sintering and SLM technology.
- Jewellery • Tools • Construction applications • Dental caps • Metal implants • Decorative models like statues, medals and keys • Military applications manufacturing automotive parts

Titanium

- Titanium is a strong, lightweight, heat- and chemical-resistant material mainly utilised in high-performance applications, including spaceships, aeroplanes, and medical devices. Tools find it extremely difficult to machine due to their tremendous strength. As a result, it is an excellent material for AM.
- •Airframe and wing structures • Compressor blades, rotors and other turbine engine components • Orthopaedic devices such as the spine, hip and knee implants • Automotive parts such as Brake callipers, brackets, wheel rims, uprights, etc

Nickel alloys

- Nickel Alloy is a precipitation-hardenable nickel-chromium alloy with significant iron, niobium, molybdenum levels and less aluminium and titanium. It has excellent corrosion resistance and strength, as well as excellent weldability and resistance to post-weld cracking.
- The DMLS (Direct et al.) or SLM method is used to 3D print Inconel nickel alloys. A laser melts an extremely tiny metal powder, layer by layer, to create a pattern. Any support structures are removed, and any finishing is completed once the design is complete. The powder that is not used is recycled and used in the next model
- • Aerospace industries • Energy industries • Process industry • Oil and gas industry • Marine applications

Aluminium & Its alloy

- Aluminium is a good material for balancing lightness and strength. It can be welded in addition to being corrosion resistant. It is less resistant to high temperatures and more susceptible to high temperatures than steel. It has been chiefly employed when a certain amount of weight is required.
- Aluminium alloys are more common than pure aluminium and have long been utilised in various industrial, aeronautical, and automotive applications. Furthermore, they have high strength-to-weight ratios and exhibit good metal fatigue and corrosion resistance
- Race cars parts • Aeronautics parts • Aerospace parts • Bicycles parts • Engine parts • Production tools

Cobalt-Chromium

- Cobalt alloys printed using the E-Beam (EBM) and DMLS procedures are higher quality than those produced using traditional methods such as lost-wax casting. Due to their stiffness, smoothness, wear-resistant surface condition, and corrosion qualities, cobalt-chromium alloys, such as Co-Cr-Mo, are frequently used to fabric prostheses in orthopaedics and deontology. In addition, several highly heat-resistant cobalt-chromium-molybdenum alloys are used in the automotive and aerospace industries.
- • Artificial joints • Dental casting • Jewellery • Gas turbines parts

Ceramics

AM technology has been successfully demonstrated its advantages in producing ceramic parts through both “direct” and “indirect” methods.

Indirect Methods

- ❖ These processes typically create a ceramic green body with a high content of organic or inorganic binders.
- ❖ Then, binder burnout and densification of the green body are conducted in a conventional sintering step.

Example: A ZrB_2 part (fuel injector strut for aircraft engine), alumina and silica cores and shells for investment casting, graphite bipolar plates for fuel cells, and bio-ceramic bone scaffolds were fabricated using SLS by laser scanning the mixture of ceramic powder and binder and then removing the binder and sintering the parts in a furnace.

Ceramics

Direct Methods

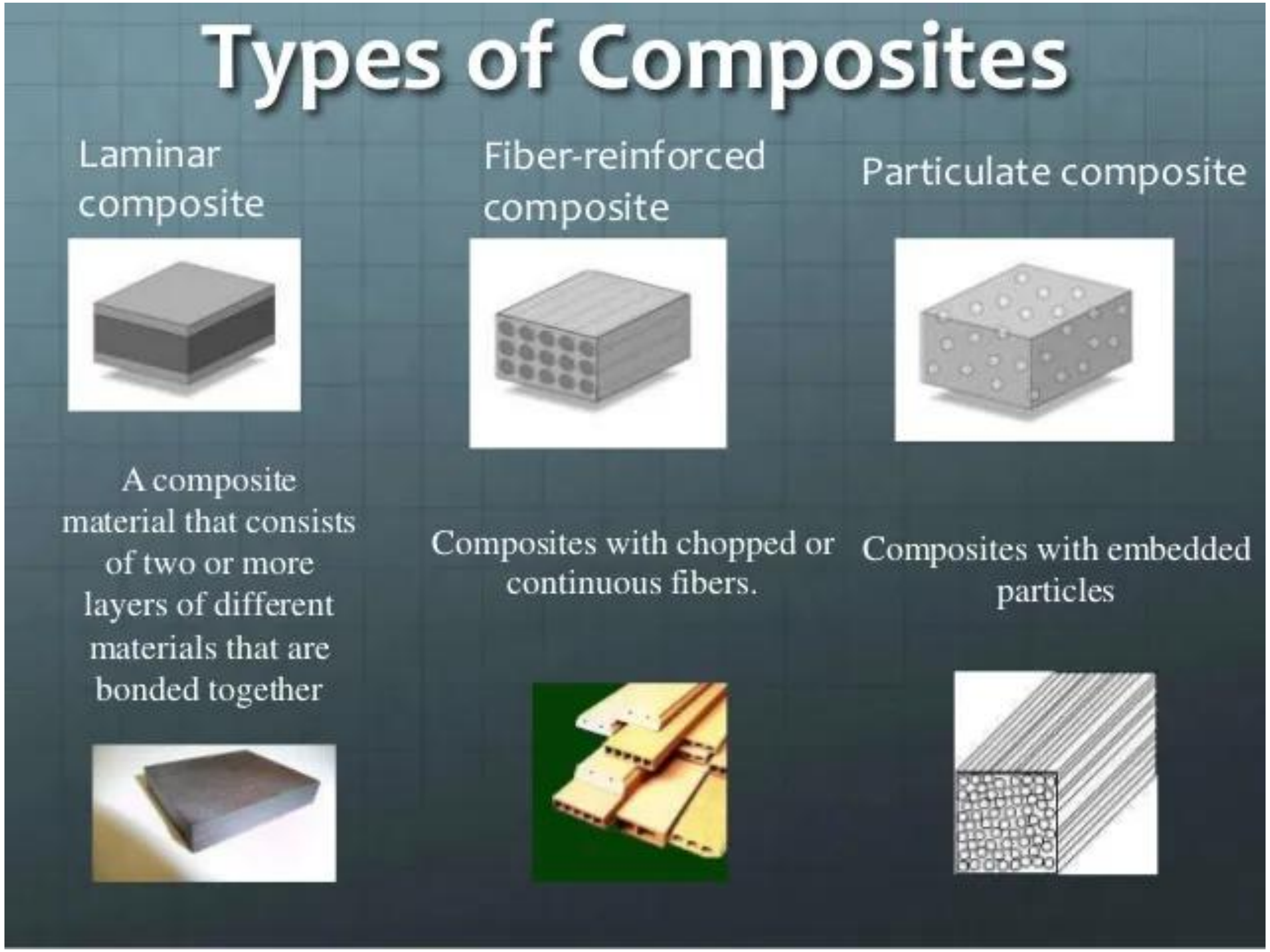
Direct fabrication of ceramic parts using AM processes is much more challenging due to the high melting temperatures of ceramics such as Al_2O_3 (> 2000°C) and SiO_2 (> 1700°C), and also the large thermal gradients, thermal stresses and residual stresses associated with melting/resolidifying in the laser based AM processes.

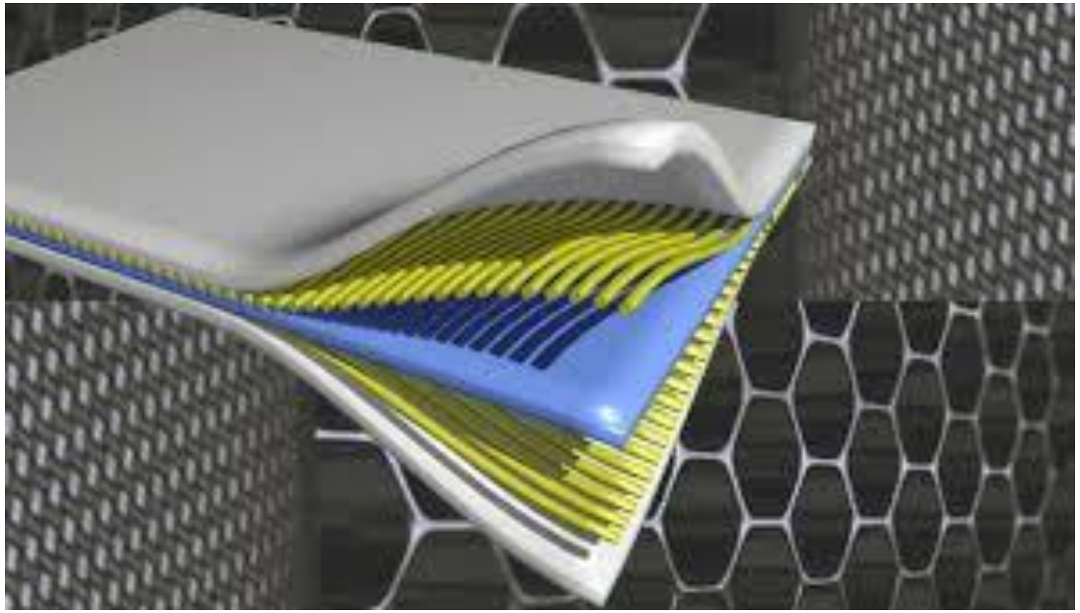
Example: SLM process was investigated to fabricate ceramic parts from a mixture of zirconia and alumina by completely melting the ceramic powder. The ceramic powder bed was preheated to a temperature higher than 1600°C to reduce thermal stresses, and nearly fully dense, crack-free parts were obtained without any post-processing. Fully dense, net-shaped, alumina parts were produced using LENS by direct laser melting of the ceramic powder.

Composites

- Composites are engineered or naturally occurring materials made from two or more constituent materials with significantly different physical or chemical properties that remain separate and distinct at the macroscopic or microscopic scale within the finished structure but exhibit properties that cannot be achieved by any of the materials acting alone.

- The materials in a composite can be mixed uniformly, resulting in a homogeneous compound (uniform composite), or non-uniformly, resulting in an inhomogeneous compound (e.g., functionally graded materials) in which the composition varies gradually over volume, leading to corresponding changes in the properties of the composite material.


Types of Composites
Laminar composite
Fiber-reinforced composite
Particulate composite
A composite material that consists of two or more layers of different materials that are bonded together
Composites with chopped or continuous fibers.
Composites with embedded particles



Composites

Uniform Composites

- Uniform composites fabricated using AM processes are usually done by employing a pre-prepared mixture of proper materials, such as a mixed powder bed for SLS, SLM and 3DP, a filament in mixed materials for FDM, a composite laminate for LOM, or a mixture of liquid photocurable resin with particulates for SLA.

- The composite materials that can be produced with AM technology include a polymer matrix, ceramic matrix, metal matrix, and fiber and particulate reinforced composites.

- Metal-metal composites (e.g., Fe-Cu and stainless steelCu), metal-ceramic composites (e.g., WC-Cu, WC-Co, WC-CuFeCo, TiC-Ni/Co/Mo, ZeB2-Cu, and TiB2-Ni), and ceramic-ceramic composites (e.g., Si-SiC) have been processed by SLS/SLM.

- These processed composites can be classified into two categories: those that aim to facilitate the process using a liquid-phase sintering mechanism, and those that combine various materials to achieve properties not possible with a single material.

Composites

Uniform Composites

- Examples of composites in the first category include Fe-Cu and stainless steel-Cu used in SLS, in which Cu acts as a binder to bond Fe or stainless steel particles rather than a reinforcement phase to enhance the mechanical or other properties of the final product.

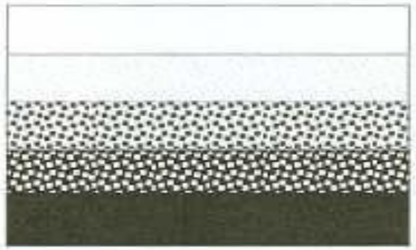
- An example of the second category is the bio-composite poly-epsilon-caprolactone and hydroxyapatite (PCL/HA) bone scaffold fabricated using SLS, with the addition of HA to enhance the strength and biocompatibility of PCL.

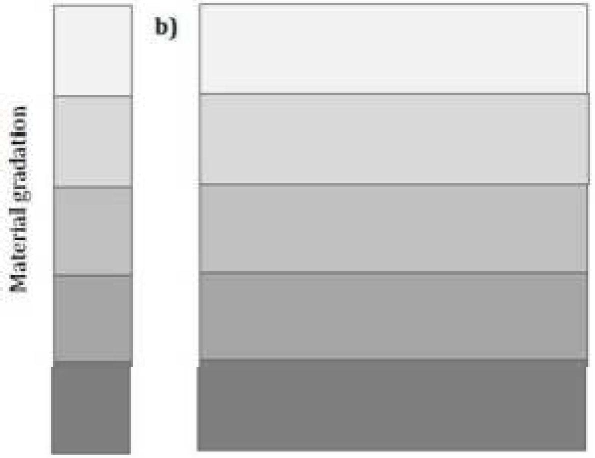
- By developing a feedstock filament with the proper composite, polymer-metal and polymer-ceramic composites could be produced with FDM. ABS-Iron composites have been made using FDM with a single-screw extruder by appropriately producing an iron particulate-filled polymeric filament. Fibers, such as short glass fibers and nanofibers (vapor-grown carbon fibers), have been added into ABS filaments to improve the mechanical properties of the parts built using FDM.

Composites: Functionally Graded Materials

Stepwise Graded Structures

An example is a spark plug which gradient is formed by changing its composition from a refractory ceramic to a metal

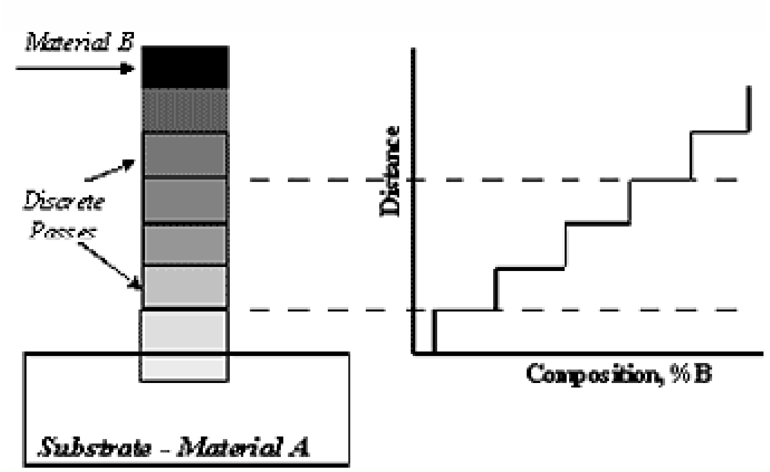


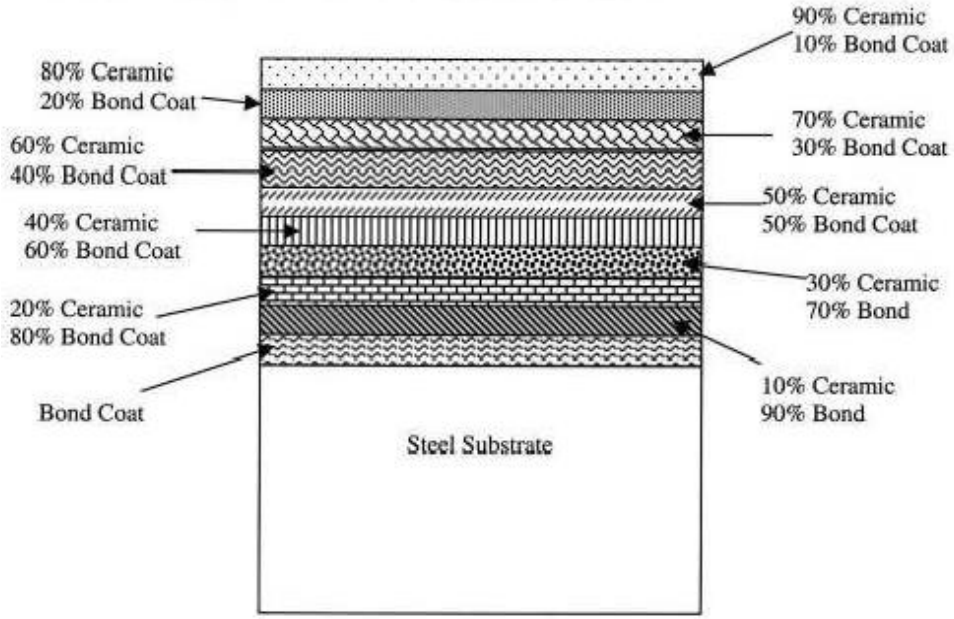


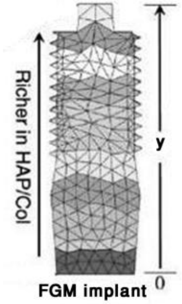
Continuous Graded Structures

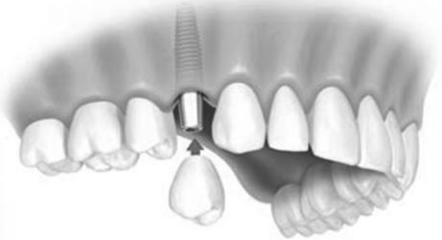
- An example is the human bone which gradient is formed by its change in porosity and composition;
- Change in porosity happens across the bone because of miniature blood vessels inside the bone.

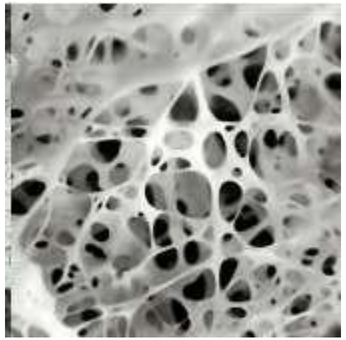
Functionally Graded Materials (FGMs)









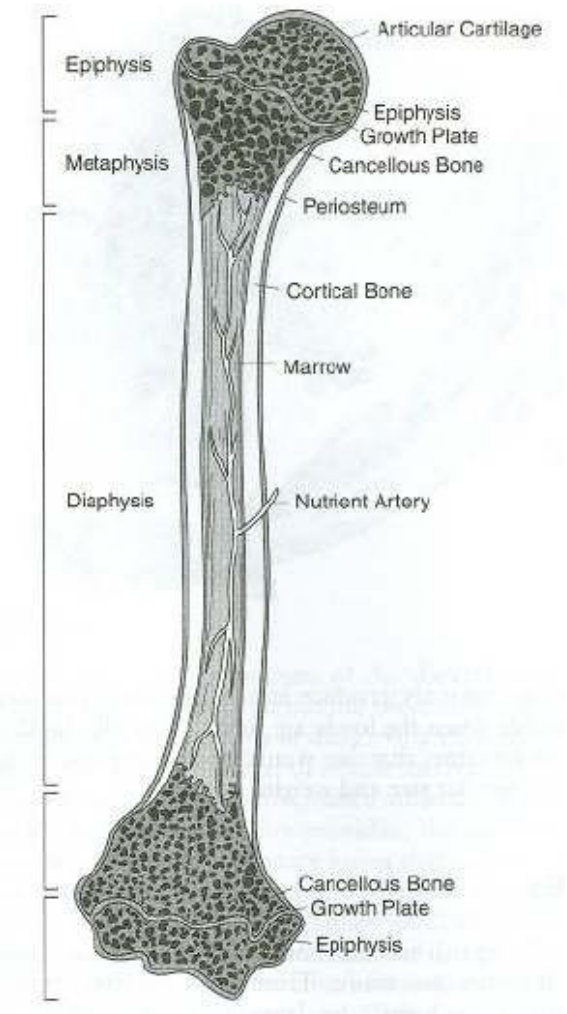


FGMs can be obtained by layered mixing of two materials of different thermo-mechanical properties with different volume ratio by gradual changing from layer to layer.

Composites: Functionally Graded Materials

The human bone is a an example of a FGM. It is a mix of collagen (ductile protein polymer) and hydroxyapatite (brittle calcium phospate ceramic).

A gradual increase in the pore distribution from the interior to the surface can pass on properties such as shock resistance, thermal insulation, catalytic efficiency, and the relaxation of the thermal stress. The distribution of the porosity affects the tensile strength and the Young’s modulus.



Composites: Functionally Graded Materials

Current applications of FGMs include:

- Structural walls that combine two or more functions including thermal and sound insulation;

- Enhanced sports equipment such as golf clubs, tennis rackets, and skis with added graded combinations of flexibility, elasticity, or rigidity;

- Enhanced body coatings for cars including graded coatings with particles such as dioxide/mica.